Design And Analysis of Algorithms
Lecture 15: Gradient Descent

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The last several lectures focused on learning from expert advice and using multiplicative-weights style algorithms to solve LPs and max flows. The rate of convergence in all of those settings had dependence on k, the number of experts, though in the non-bandit setting, where we observe the loss/gain of every expert regardless of our choice in round t, this dependence was only logarithmic. In this lecture, we show it's possible to learn over an uncountable number of experts (for example, each point in the $\mathbb{R}^2$ plane might represent an expert) with polynomial rates of convergence *if we assume some structure connecting the payoffs between different experts.*

The sorts of structure we'll investigate today are: (1) G-Lipschitzness, which says the loss between to nearby points x, y can differ by at most $G||x-y||_2$ (thus having an accurate estimate of the losses at x gives reasonable estimates of nearby y), (2) that *some* function of $x \in \mathbb{R}^2$ governs the payoffs, (3) that this loss function is convex, and (4) stronger versions of convexity and its dual, called smoothness, which lower and upper bound the rate of change of the loss function, respectively.

It is natural to ask what type of access we have to the losses for each action; the most direct analog to learning from expert advice would be to give our algorithm (query or some other) access to the loss of any point in the plane regardless of its choice in round t. Then, there would be two goals for our algorithm: how to learn a policy which quickly begins to pick points with small loss, and how to query these losses some small number of times. Instead, we turn these two knobs into one and say we only observe the loss for the action we choose at round t; thus, the rate and number of queries we must make to the loss function are one and the same. We will also assume we get a directional "hint" when we query action x, in the form of the gradient of ℓ evaluated at x. This connects this setting more closely to bandit learning, where one only observes the payoff from the expert selected in round t, with the added help of the gradient information.

We now introduce the formal notation we will use for this lecture. Let $\mathcal{X}$ be the set of available actions, and $\ell : \mathcal{X} \to \mathbb{R}$ represent the *loss* function we face. We will assume throughout that every loss function we face is convex, which we define below.

Definition 1 (Convexity)**.** *A function ℓ is convex if, for every $x, y \in \mathcal{X}$, and every $\lambda \in [0,1]$, we have that*

$$\lambda\ell(x) + (1-\lambda)\ell(y) \geq \ell(\lambda x + (1-\lambda)y).$$

We also will need that the loss function does not change too rapidly anywhere:

Definition 2 (G-Lipschitzness)**.** *A function ℓ is G-Lipschitz, if for all $x, y, \in X$,*

$$|\ell(x) - \ell(y)| \leq G \cdot ||x-y||_2.$$

The following equivalences will be useful for our analysis. Recall that $\nabla\ell$ refers to the gradient of ℓ, or the coordinate-wise derivative. The first state that convexity is related to the gradient of a differentiable function.

Remark 1. *If ℓ is differentiable, it is convex iff for all $x, y \in \mathcal{X}$, $\ell(y) - \ell(x) \geq \langle \nabla\ell(x), y - x\rangle$.*

The next states that the gradient of a differentiable function also governs whether or not the function is Lipschitz.

Remark 2. *If ℓ is differentiable, then it is G-Lipschitz iff $||\nabla \ell(x)||_2 \leq G$ for all $x \in \mathcal{X}$.*

Recall that $\nabla^2 \ell$ is the Hessian of ℓ, such that $(\nabla^2 \ell)_{i,j} = \frac{\partial^2 \ell}{\partial x_i \partial x_j}$. This last remark states that convexity is equivalent to having a PSD Hessian.

Remark 3. *If ℓ is twice-differentiable, it is convex iff $\nabla^2 \ell \succeq 0$.*[1]

we will study two versions of optimizing over an unknown loss function ℓ, that where $\mathcal{X}$ is all of $\mathbb{R}^n$ and that where $\mathcal{X}$ is some convex subset of $\mathbb{R}^n$:

- **Unconstrained Convex Minimization** Given convex ℓ, find $\min_{x \in \mathbb{R}^n} \ell(x)$.
- **Constrained Convex Minimization** Given a convex ℓ and convex set $\mathcal{X}$, find $|min_{x \in \mathcal{X}} \ell(x)$.

0.1 The Unconstrained Optimization Problem

The first version we consider allows us to pick any point in $\mathbb{R}^n$ as our action in round t. So, finding $\min_{x \in \mathbb{R}^n} \ell(x)$ is equivalent to finding some global minimum for ℓ, and since ℓ is convex, this is equivalent to finding some x^* for which

$$\nabla \ell(x^*) = 0.$$

Our algorithm for finding a sequence of points which approach x^* is as follows. When we pick a point x^t, we observe $\nabla \ell(x^t)$, the rate of change of ℓ in all directions around x^t. So, we move in the direction of greatest decrease of ℓ, namely in the direction of $-\nabla \ell(x^t)$. We will take a step of size η^t, which we will define later to optimize the rate at which our algorithm approaches x^*.

Algorithm 1: (Unconstrained) Gradient Descent

```
x_0 ← starting point;
for t = 1 to T do
    x^{t+1} ← x^t − η^t ∇ℓ(x^t);
end
Return x̂ = (1/T) Σ_{t=1}^{T} x^t
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Gradient Descent performs quite well both in this and other settings; the following theorem guides how to pick the step-size η^t to have fast convergence.

Theorem 1. *Suppose ℓ is convex and G-Lipschitz, and $||x_0 - x^*||_2 \leq D$. Then, if $\eta^t = \eta = \frac{\epsilon}{G^2}$, and $T \geq \left(\frac{GD}{\epsilon}\right)^2$, we have $\ell(\hat{x}) - \ell(x^*) \leq \epsilon$.*

This proof is rather straightforward: we keep track of how much progress we must make between rounds by looking at $||x^t - x^{t+1}||$. We use the Lipschitzness of ℓ to argue about the change in loss function which might occur by taking a step which could be too large, and the fact that the step was taken in the direction of the gradient to imply that the change will be large enough to make progress.

[1] This notation refers to the matrix being positive semi-definite.

Proof. Define $\Phi^t = \frac{1}{2\eta}||x^t - x^*||_2^2$; this measures the closeness of our action in round t to the optimal action. Now notice that

$$\begin{aligned}
\Phi^{t+1} - \Phi^t &= \frac{1}{2\eta}\left(||x^{t+1} - x^*||_2^2 - ||x^t - x^*||_2^2\right) \\
&= \frac{1}{2\eta}\left(||x^{t+1} - x^t + x^t - x^*||_2^2 - ||x^t - x^*||_2^2\right) \\
&= \frac{1}{2\eta}\left(2\langle x^{t+1} - x^t, x^t - x^*\rangle + ||x^{t+1} - x^t||_2^2\right) \\
&= \frac{1}{2\eta}\left(2\langle -\eta\nabla\ell(x^t), x^t - x^*\rangle + ||-\eta\nabla\ell(x^t)||_2^2\right) \\
&\leq \frac{1}{2\eta}\left(2\eta\left(\ell(x^*) - \ell(x^t)\right) + \eta^2 G^2\right) \\
&= \ell(x^*) - \ell(x^t) + \frac{\eta}{2}G^2
\end{aligned}$$

where in the second-to-last line we used the first-order condition (Remark 1) of convexity and the upper bound on the derivative of any Lipschitz function (Remark 2).

Now, we bound the term we actually care about: the difference in loss between our average point and that of the optimal point:

$$\begin{aligned}
\ell(\hat{x}) - \ell(x^*) &= \ell(\frac{1}{T}\sum_t x^t) - \ell(x^*) \\
&\leq \frac{1}{T}\sum_t \ell(x^t) - \ell(x^*) && \text{By convexity of } \ell \\
&\leq \frac{1}{T}\sum(\Phi^t - \Phi^{t+1} + \frac{\eta G^2}{2}) && \text{By the derivation above} \\
&= \frac{1}{T}\left(\Phi^0 - \Phi^T\right) + \frac{\eta G^2}{2} \\
&\leq \frac{1}{T}\left(\Phi^0\right) + \frac{\eta G^2}{2} \\
&\leq \frac{1}{2\eta T}\left(||x^0 - x^*||_2^2\right) + \frac{\eta G^2}{2} \\
&\leq \frac{D^2}{2\eta T} + \frac{\eta G^2}{2} && \text{By assumption regarding } x^0 \text{ .}
\end{aligned}$$

Substitution of T and η as defined in the theorem completes the proof. □

Without more assumptions on ℓ, the dependence of the rate on ϵ is tight. We will show later that additional assumptions can improve this guarantee.

0.2 The Constrained Problem

We now focus on the constrained problem of finding the minimum of ℓ in some convex space $\mathcal{X}$: this problem is harder for two reasons. One, taking a step in the direction of the gradient might take you from inside of $\mathcal{X}$ to outside of $\mathcal{X}$. Second, the global minimum of ℓ might not belong to $\mathcal{X}$, and so there may be no point in $\mathcal{X}$ with gradient zero. Nevertheless, a small variant of gradient descent is still a good algorithm for this problem; it is still a good idea to follow the gradient of the loss function.

Let $\Pi_{\mathcal{X}}(x) = \arg\min_{y\in\mathcal{X}} ||x - y||_2$ denote the function which projects some point to the nearest point in some set $\mathcal{X}$. Then, we can project any point outside of $\mathcal{X}$ back into $\mathcal{X}$ while moving as little as possible.

Algorithm 2: Projected Gradient Descent

$x_0 \leftarrow$ starting point;
for $t = 1$ *to* T **do**
  $\tilde{x}^{t+1} \leftarrow x^t - \eta^t \nabla \ell(x^t)$;
  $x^{t+1} \leftarrow \Pi_{\mathcal{X}}(\tilde{x}^{t+1})$;
end
Return $\hat{x} = \frac{1}{T}\sum_{t=1}^{T} x^t$

Remark 4. *If $\mathcal{X}$ is defined as an intersection of halfspaces, $\Pi_{\mathcal{X}}$ can be computed by iteratively finding the nearest point on each halfspace and then taking the point which minimizes this distance. For more general convex $\mathcal{X}$, this might be computationally more expensive.*

This slight modification to the algorithm ensures that it never chooses an infeasible point, and admits nearly the same analysis of its rate.

Theorem 2. *Suppose $\mathcal{X}$ is convex with diameter D, and ℓ is convex and G-Lipschitz. If $x^0 \in \mathcal{X}$, $T = \left(\frac{GD}{\epsilon}\right)^2, \eta = \frac{\epsilon}{G^2}$, then $\ell(\hat{x}) - \ell(x^*) \leq \epsilon$.*

Proof. The only difference in the argument is that $\eta \nabla \ell(x^t) = \tilde{x}^{t+1} - x^t$, rather than the same statement without the $\tilde{}$. So, if one defines $\Phi^t = \frac{1}{2\eta}||\tilde{x}^t - x^*||_2^2$, then the only line which changes from the previous proof is that the line which was previously

$$\frac{1}{2\eta}\left(2\langle x^{t+1} - x^t, x^t - x^*\rangle + ||x^{t+1} - x^t|_2^2\right)$$

becomes

$$\frac{1}{2\eta}\left(2\langle \tilde{x}^{t+1} - x^t, x^t - x^*\rangle + ||\tilde{x}^{t+1} - x^t|_2^2 + ||x^t - x^*||_2^2 - ||\tilde{x}^t - x^*||_2^2\right)$$

and the last two terms sum to at most 0, since any point projected to the closest point in a convex set is closer to every point in that convex set than the original point. □

0.3 Generalizations: Varying Loss Functions and subgradients

Gradient Descent and its analysis are very general: so general, we can also use it when facing (convex, Lipschitz) loss functions which differ every day ℓ^t rather than a fixed ℓ. The only change to the algorithm will be to take a step at time t in the direction $-\eta \nabla \ell^t(x^t)$, the gradient of the loss function at time t.

Similarly, we can generalize the problem where the loss functions don't necessarily have well-defined gradients, but we do have *subgradients* for them:

Definition 3. *A vector z_x is a subgradient fo ℓ at point x if*

$$f(y) - f(x) \geq \langle z_x, y - x\rangle \qquad \forall y \in \mathbb{R}^n.$$

Then, we just need to use the subgradients rather than the true gradients to define the algorithm and analyze it.

0.4 Additional Assumptions

Gradient descent's performance as outlined above is (essentially) optimal in terms of the rate's dependence on ϵ of $\frac{1}{\epsilon^2}$, if we don't make more assumptions about the structure of the loss function(s). If we have lower and upper bounds on the "amount" of convexity the loss functions much have (namely, an upper and lower bound on the slack in the convexity constraint), we can get tighter convergence guarantees.

Definition 4 (α-strong convexity)**.** *A function ℓ is α-strongly convex if, for all $x, y \in \mathcal{X}$, we have*

$$\ell(y) - \ell(x) \geq \nabla \ell(x)^T (y - x) + \frac{\alpha}{2} ||y - x||_2^2.$$

Informally, α can be thought of a lower bound on the amount of curvature the loss function. Smoothness, defined below, is just the opposite: it is an upper bound on the curvature of the loss function.

Definition 5 (β-smoothness)**.** *A function ℓ is β-smooth if, for all $x, y \in \mathcal{X}$, we have*

$$\ell(y) - \ell(x) \leq \nabla \ell(x)^T (y - x) + \frac{\beta}{2} ||y - x||_2^2.$$

These two properties together give us a good understanding of how the loss function might change between two adjacent points: it changes by some amount bounded above and below by linear functions of $||x - y||_2^2$.

Definition 6 ($\kappa = \frac{\alpha}{\beta}$-conditioned)**.** *A function is called $\kappa = \frac{\alpha}{\beta}$-well conditioned if it is α-strongly convex and β-smooth.*

When a function is well-conditioned, we can set the step size as a function of α and β to get faster convergence, as the following theorem suggests.

Theorem 3. *Suppose ℓ is $\frac{\alpha}{\beta}$-well conditioned. Then, if we set $\eta = \frac{1}{\beta}$, we have that the unconstrained gradient descent algorithm satisfies*

$$\ell(x^t) - \ell(x^*) \leq \frac{\beta}{2} \exp\left(\frac{-t}{k}\right) ||x^0 - x^*||^2.$$

Proof. We can use strong convexity to imply

$$\begin{aligned}
\ell(x^t) - \ell(x^*) &\leq \langle \nabla \ell^t(x^t), x^t - x^* \rangle - \frac{\alpha}{2} ||x^t - x^*||^2 \\
&\leq ||\nabla \ell^t(x^t)|| ||x^t - x^*|| - \frac{\alpha}{2} ||x^t - x^*||^2 \\
&\leq \frac{1}{2\alpha} ||\nabla \ell^t(x^t)||^2 && \text{since this is maximized when } ||x^t - x^*|| = \frac{||\nabla \ell^t(x^t)||}{\alpha}
\end{aligned}$$

and β-smoothness to imply

$$\begin{aligned}
\ell(x^{t+1}) - \ell(x^t) &\leq -\eta ||\nabla \ell^t(x^t)||^2 + \eta^2 \frac{\beta}{2} ||\nabla \ell^t(x^t)||^2 \\
&\leq \frac{-1}{2\beta} ||\nabla \ell^t(x^t)||^2 && \text{Substituting in } \eta
\end{aligned}$$

which together imply that

$$\ell(x^{t+1}) - \ell(x^t) \leq \frac{\alpha}{\beta} \left(\ell(x^t) - \ell(x^*) \right).$$

Using the above iteratively, we know that

$$\ell(x^{t+1}) - \ell(x^*) \leq (1 - \frac{\alpha}{\beta})(\ell(x^t) - \ell(x^*)) \leq (1 - \frac{1}{\kappa})^t (\ell(x^0) - \ell(x^*))$$

which can be converted into the theorem statement because the loss function is β-smooth, so $\ell(x^0) - \ell(x^*) \leq \frac{\beta}{2} ||x^0 - x^*||^2$. $\square$

1 Comparison with MW

We briefly now mention that we could use gradient descent in a setting like learning from n discrete experts, and analyze the regret of doing so. In that case, $K = -[-1, 1]^n$, $\ell^t \in [0, 1]^n$, so to achieve error at most ϵ we need $T \geq \left(\frac{DG}{\epsilon}\right)^2$ or $T \geq \frac{2N}{\epsilon^2}$, since $||\nabla \ell_i||_2 \leq \sqrt{n}$, whereas MW achieved dependence only logarithmic in n. The next lecture will show that this is because we're doing updates with respect to ℓ_2 rather than ℓ_1, and that doing them with respect to ℓ_1 will improve this dependence on n.